

Master MVA - ENS Paris-Saclay

Statistics Refresher - Tutorial

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1 Empirical frequencies and categorical data

Let $X_1, \dots, X_n$ be an i.i.d. sample taking values in $\{1, \dots, K\}$, with

$$\mathbb{P}(X_i = k) = p_k, \quad p_k \geq 0, \quad \sum_{k=1}^K p_k = 1.$$

For each observation, define its *one-hot encoding*

$$Z_i = (\mathbb{1}_{\{X_i=1\}}, \dots, \mathbb{1}_{\{X_i=K\}})^\top$$

and the empirical frequency vector

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n Z_i.$$

1. Interpret $\hat{p}_k$. Show that $\hat{p}$ is a probability vector (the sum of its coordinates is 1).
2. Compute $\mathbb{E}[\hat{p}_k]$, $\text{Var}(\hat{p}_k)$ and, for $k \neq \ell$, $\text{Cov}(\hat{p}_k, \hat{p}_\ell)$. Interpret the sign of this covariance.
3. Show that $\hat{p}$ converges almost surely to $p = (p_1, \dots, p_K)^\top$.
4. A finite *mixture model* can be described by introducing a latent variable $C \in \{1, \dots, K\}$ such that

$$\mathbb{P}(C = k) = \pi_k, \quad X \mid (C = k) \sim P_k.$$

Equivalently,

$$P_X = \sum_{k=1}^K \pi_k P_k.$$

In a Gaussian mixture model, $P_k = \mathcal{N}(\mu_k, \Sigma_k)$. For an i.i.d. sample, define

$$U_{ik} = \mathbb{1}_{\{C_i=k\}}.$$

Interpret U_{ik} and $N_k = \sum_{i=1}^n U_{ik}$. What changes in practice when the variables C_i are not observed?

5. Identify the distribution of

$$N = (N_1, \dots, N_K)$$

when the labels C_i are observed and independent.

Exercise 1: Empirical frequencies and categorical data

1) $\forall k, \hat{p}_k = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{X_i=k\}}$ proportion of observations of class k

$\sum_k \hat{p}_k = \frac{1}{n} \sum_{k,i} \mathbb{1}_{\{X_i=k\}} = 1$ because each X_i is in 1 category

2) $\hat{p}_k \sim B(n, p_k)$ (category k or not?)

$$E[\hat{p}_k] = p_k$$

$$\text{Var}(\hat{p}_k) = \frac{p_k(1-p_k)}{n}$$

$$k \neq l: \text{Cov}(\mathbb{1}_{\{X_i=k\}}, \mathbb{1}_{\{X_i=l\}}) = E[\underbrace{\mathbb{1}_{\{X_i=k\}} \mathbb{1}_{\{X_i=l\}}}_{=0 \text{ since } k \neq l}] - p_k p_l = -p_k p_l$$

$$\text{Cov}(\hat{p}_k, \hat{p}_l) = -\frac{p_k p_l}{n}$$

3) LLN

Law of large numbers (L^2):

If $X_i \in L^2$ and $E[X_i] = m$, $\frac{S_n}{n} \xrightarrow{L^2} m$



2 Bregman divergence, KL divergence and cross-entropy

Let $C \subset \mathbb{R}^d$ be a convex set and let $F : C \rightarrow \mathbb{R}$ be continuously differentiable and strictly convex. The *Bregman divergence* associated with F is

$$D_F(p, q) = F(p) - F(q) - \langle \nabla F(q), p - q \rangle.$$

1. Show that if $F(x) = \|x\|_2^2$, then

$$D_F(p, q) = \|p - q\|_2^2.$$

2. Consider two probability vectors $p = (p_i)_{1 \leq i \leq K}$ and $q = (q_i)_{1 \leq i \leq K}$ with strictly positive coordinates. Define

$$H(p) = -\sum_{i=1}^K p_i \log(p_i), \quad KL(p, q) = \sum_{i=1}^K p_i \log\left(\frac{p_i}{q_i}\right).$$

Show that $KL(p, q)$ is the Bregman divergence associated with $-H$.

3. Show that $KL(p, q) \geq 0$, with equality if and only if $p = q$.

Remark: KL is not a distance: it is not symmetric and does not satisfy the triangle inequality.

4. Let $R : \mathcal{W} \rightarrow \mathbb{R}$ be a differentiable convex function on an open convex set $\mathcal{W} \subset \mathbb{R}^d$, and let w^* be a minimizer of R . Define

$$\mathcal{E}(w) = R(w) - R(w^*).$$

Show that

$$\mathcal{E}(w) = D_R(w, w^*).$$

Which first-order optimality condition is used?

5. A *probabilistic classifier* for K classes outputs a probability vector

$$q = (q_1, \dots, q_K), \quad q_k > 0, \quad \sum_{k=1}^K q_k = 1,$$

where q_k is interpreted as the predicted probability of class k . If the true class is $Y \in \{1, \dots, K\}$, define the *log-loss* or *cross-entropy loss* by

$$l(q, Y) = -\log(q_Y).$$

Assume that the true distribution of Y is p and define

$$R(q) = \mathbb{E}[l(q, Y)].$$

Show that

$$R(q) = H(p) + KL(p, q).$$

Deduce the population-optimal prediction and identify the excess risk.

3 PCA

Let $x_1, \dots, x_n \in \mathbb{R}^p$ and

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The empirical covariance matrix is

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^\top.$$

We assume that the observations are centered, i.e. $\bar{x} = 0$. Let

$$\lambda_1 \geq \dots \geq \lambda_p \geq 0$$

be the eigenvalues of $\hat{\Sigma}$, with associated orthonormal eigenvectors $v_1, \dots, v_p$.

Exercice 2: Bregman divergence, KL divergence and cross-entropy

1) $D_F(p, q) = F(p) - F(q) - \langle \nabla F(q), p - q \rangle$

$F(x) = \|x\|_2^2 \quad \nabla F(x) = 2x$

$$\begin{aligned} D_F(p, q) &= \|p\|_2^2 - \|q\|_2^2 - \langle 2q, p - q \rangle \\ &= \|p\|_2^2 - \|q\|_2^2 - 2\langle q, p \rangle + 2\|q\|_2^2 \\ &= \|p\|_2^2 + \|q\|_2^2 - 2\langle q, p \rangle \\ &= \|p - q\|_2^2 \end{aligned}$$

2) $KL(p, q) = \sum_{i=1}^K p_i \log\left(\frac{p_i}{q_i}\right) \quad H(p) = - \sum_{i=1}^K p_i \log(p_i)$

$\nabla(-H)(q)_i = \frac{\partial F}{\partial q_i}(q) = 1 + \log(q_i)$

$$\begin{aligned} D_{-H}(p, q) &= -H(p) + H(q) + \langle \nabla H(q), p - q \rangle \\ &= \sum_{i=1}^K (p_i \log(p_i) - q_i \log(q_i) - (p_i - q_i)(1 + \log(q_i))) \end{aligned}$$

$$= \sum (p_i \log(p_i) - p_i + q_i - q_i \log(q_i))$$

$$= KL(p, q) + \cancel{\sum q_i} - \cancel{\sum p_i} \quad \text{direction}$$

3) By convexity, $\forall p, q, \quad \underbrace{F(p) = F(q) + \langle \nabla F(q), p - q \rangle}_{\text{formule de la tangente}}$

$$\Leftrightarrow D_{KL}(p, q) \geq 0$$

with equality iff $p = q$ by strict convexity

4) w^* is a minimum of R so $\nabla R(w^*) = 0$

$$\text{So } \mathcal{E}(w) = R(w) - R(w^*) = D_R(w)$$

5) $y \sim p$

$$l(q, y) = -\log(q_y)$$

$$R(y) = E_p(-\log q_y) = \sum_{i=1}^K p_i \log q_i + \sum p_i \log p_i - \underbrace{p_i \log p_i}_H$$

$$= H(p) + D_{KL}(p, q)$$

$$q^* = p \quad R(q^*) = H(p)$$

$$\mathcal{E}(q) = D_{KL}(p, q)$$

1. Show that

$$v_1 \in \operatorname{argmax}_{\|v\|_2=1} \frac{1}{n} \sum_{i=1}^n (v^\top x_i)^2.$$

2. Explain how to obtain the first k principal directions.
3. The total empirical variance is defined as $\operatorname{tr}(\hat{\Sigma})$. Show that the fraction of variance explained by the first k principal components is

$$\frac{\lambda_1 + \dots + \lambda_k}{\lambda_1 + \dots + \lambda_p}.$$

4. Let $V \subset \mathbb{R}^p$ be a k -dimensional subspace and let P_V denote the orthogonal projection onto V . Show that the PCA subspace also solves

$$\operatorname{argmin}_{\dim(V)=k} \frac{1}{n} \sum_{i=1}^n \|x_i - P_V x_i\|_2^2.$$

5. Let $X \in \mathbb{R}^{n \times p}$ be the centered data matrix whose i^{th} row is $x_i^\top$. A singular value decomposition (SVD) of X is a factorization

$$X = USV^\top,$$

where U and V have orthonormal columns and S is diagonal with non-negative diagonal entries. Relate the principal directions to this SVD.

4 Method of moments vs maximum likelihood estimation

Recall that $Beta(\alpha, \beta)$, $\alpha, \beta > 0$, is the distribution on $[0, 1]$ with density

$$p_{\alpha, \beta}(x) \propto x^{\alpha-1} (1-x)^{\beta-1}.$$

Its mean and variance are respectively

$$\frac{\alpha}{\alpha + \beta} \quad \text{and} \quad \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

We consider the statistical model consisting of the uniform distributions on $[0, \theta]$, $\theta > 0$, and estimate θ from an i.i.d. sample $X_1, \dots, X_n$.

1. Compute the moment estimator $\hat{\theta}_{MO}$.
2. Show that the maximum likelihood estimator $\hat{\theta}_{MLE}$ exists and is unique and compute it.
3. Show that

$$\frac{\hat{\theta}_{MLE}}{\theta} \sim Beta(n, 1).$$

Deduce the bias and variance of $\hat{\theta}_{MLE}$.

4. Compute the bias and variance of $\hat{\theta}_{MO}$. Compare

$$MSE_\theta(\hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \theta)^2]$$

for the two estimators.

5. Construct an unbiased estimator $\tilde{\theta}$ from $\hat{\theta}_{MLE}$. Compute its MSE and compare it with that of $\hat{\theta}_{MO}$.
6. Show that

$$n \left(1 - \frac{\hat{\theta}_{MLE}}{\theta} \right) \xrightarrow{\mathcal{L}} \mathcal{E}(1).$$

Which usual regularity assumption of the standard asymptotic theory of maximum likelihood estimators fails in this model?

Exercise 3: PCA

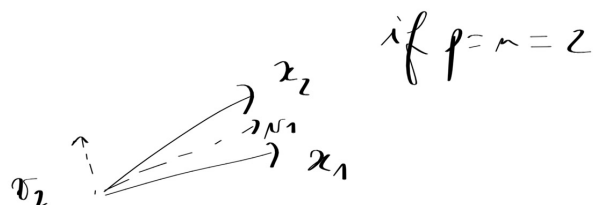
1) Let $v \in \mathbb{R}^p$, $\|v\|^2 = 1$

$$v = \sum_{i=1}^p a_i v_i \quad \|v\|^2 = 1 = \sum_{i=1}^p a_i^2$$

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n v^T x_i x_i^T v &= v^T \hat{\Sigma} v \\ &= v^T \sum_{i=1}^p \lambda_i a_i v_i \\ &= \sum_{i=1}^p \lambda_i a_i^2 \leq \lambda_1 \\ &\quad \underbrace{\leq \lambda_1} \end{aligned}$$

$$v_1^T \hat{\Sigma} v_1 = \lambda_1$$

So $v_1 \in \operatorname{argmax}$



2) Take $v_1, \dots, v_k$

3) Empirical variance $\operatorname{Tr}(\hat{\Sigma}) = \sum_{i=1}^p \lambda_i$

Projection on $\operatorname{Span}(v_1, \dots, v_k)$

$$\hookrightarrow \operatorname{Var}(\hat{\Sigma}_k) = \sum_{i=1}^k \lambda_i$$

4) $\|x - p_v(x)\|^2 = \|x\|^2 - \|p_v(x)\|^2$ (th. Pythagore)

5) If X is not a square matrix

Solution: $\frac{1}{n} X X^T \in \mathcal{S}_n^+(\mathbb{R})$

Exercise 4: method of moments VS maximum likelihood estimation

1) $\frac{1}{n} \sum_{i=1}^n X_i = \frac{\hat{\theta}_{MO}}{2}$ car $E(X) = \frac{0+\theta}{2}$ $\swarrow$ loi uniforme $U([0, \theta])$

$\hat{\theta}_{MO} = \frac{2}{n} \sum_{i=1}^n X_i$ $X_1, \dots, X_n \sim U([0, \theta])_{iid}$

$$E_{\theta}[\hat{\theta}_{MO}] = \frac{2}{n} \sum E_{\theta}[X_i] = \theta$$

$\underbrace{\qquad\qquad\qquad}_{\frac{\theta}{2}}$

$$So \text{ } b_{\theta}(\hat{\theta}_{MO}) = 0$$

$$Var_{\theta}(\hat{\theta}_{MO}) = \frac{4}{n^2} \sum Var(X_i) = \frac{\theta^2}{3n} = R(0, \hat{\theta}_{MO})$$

$\underbrace{\qquad\qquad\qquad}_{\frac{\theta^2}{12}}$

2) definition: MLE maximum likelihood estimator

$$\forall z \in \mathcal{Z} \quad MLE = \underset{\theta \in \Theta}{\operatorname{argmax}} L(z, \theta) \quad \text{if it exists}$$

definition: Likelihood function

$$L: \begin{cases} \mathcal{Z} \times \Theta \rightarrow \mathbb{R}^+ \\ (z, \theta) \mapsto p_{\theta}(z) \end{cases}$$

$$L(z, \theta) = \prod_{i=1}^n \frac{1}{\theta} \mathbb{1}_{\{X_i \in [0, \theta]\}} = \frac{1}{\theta^n} \prod_{i=1}^n \mathbb{1}_{\{X_i \in [0, \theta]\}} = \prod_{i=1}^n p_{\theta}(x_i)$$

(If $z = (x_1, \dots, x_n)$ iid, $L(z, \theta) = \prod_{i=1}^n p_{\theta}(x_i)$)

$$\hat{\theta}_{MLE} = \max (X_i)$$

3) $\forall x \in [0, 1]$.

$$P\left(\frac{\hat{\theta}_{MLE}}{\theta} \leq x\right) = P\left(\frac{\max(X_i)}{\theta} \leq x\right) \stackrel{\text{ind}}{=} P\left(\bigcap_{i=1}^n \left(\frac{X_i}{\theta} \leq x\right)\right)$$

$$\stackrel{\substack{\text{dérivée} \\ \text{fn repartition}}}{=} F_{U[0,1]}(x)^n = \left(\int_0^x dt\right)^n = x^n$$

$$P_{\frac{\hat{\theta}}{\theta}}(x) = nx^{n-1} \mathbb{1}_{\{0 \leq x \leq 1\}} \quad (\text{densité})$$

$$\frac{\hat{\theta}_{MLE}}{\theta} \sim \text{Beta}(n, 1) \quad (\text{m densité avec } \alpha=n \text{ et } \beta=1 \rightarrow \text{voir énoncé})$$

$$E[\hat{\theta}_{MLE}] = \theta \times \frac{n}{n+1}$$

$$b_0(\hat{\theta}_{MLE}) = -\frac{\theta}{n+1}$$

$$\text{Var}(\hat{\theta}_{MLE}) = \frac{n}{(n+1)^2(n+2)} \theta^2$$

$$\text{MSE}(\hat{\theta}_{MLE}) = \frac{\theta^2}{(n+1)^2} + \frac{n}{(n+1)^2(n+2)} \theta^2 = \frac{2\theta^2}{(n+1)(n+2)}$$

$$(\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\overset{\text{valeurs réelles}}{\hat{y}_i} - \underset{\text{valeurs prédites}}{\bar{y}_i})^2) \quad (\text{risk} \downarrow \text{like } \frac{1}{n^2})$$

$$(\text{MSE} = \text{var}(\hat{\theta}) + \text{biais}(\hat{\theta})^2)$$

estimator asymptotically unbiased

$$5) E[\hat{\theta}_{MLE}] = \frac{n\theta}{n+1} \quad E\left[\underbrace{\frac{n+1}{n} \hat{\theta}_{MLE}}_{\text{unbiased estimator}}\right] = \theta$$

$$6) P\left(\left(1 - \frac{\hat{\theta}_{MLE}}{\theta}\right)^n > x\right) = P\left(\frac{\hat{\theta}_{MLE}}{\theta} \leq 1 - \frac{x}{n}\right) = \left(1 - \frac{x}{n}\right)^n \xrightarrow[n \rightarrow \infty]{} e^{-x}$$

$E[\hat{y}] = y$

$$X \sim \mathcal{E}(\lambda) \Rightarrow f_X(x) = (\lambda - e^{-x}) 1_{[0, \lambda]}(x)$$

$\uparrow$
 loi exp.

$\text{supp } p_\theta = [0, \theta]$ depends on θ (assumption ~~faute~~)

L'hypothèse de régularité qui échoue est l'hypothèse de support fixe : le support de la densité dépend de θ , puisque $\text{Supp}(X)=[0, \theta]$.

Cette dépendance du support entraîne notamment l'échec des conditions classiques de différentiabilité permettant d'obtenir la théorie asymptotique normale standard du MLE.

5 Maximum likelihood estimators

- Let $X_1, \dots, X_n$ be i.i.d. $Ber(p)$ random variables, $p \in [0, 1]$.
 - Compute the MLE $\hat{p}_{MLE}$, including the cases where all observations are equal to 0 or all are equal to 1.
 - Assume now that the true parameter satisfies $p \in (0, 1)$. Compute the Fisher information for one observation and for the sample.
 - Compute the asymptotic distribution of

$$\sqrt{n}(\hat{p}_{MLE} - p).$$
 Compare its asymptotic variance with the Cramér–Rao bound.
 - Using the asymptotic normality above, construct the usual Wald confidence interval at level 95%. Why can this interval be problematic when p is close to 0 or 1?
- Let $X_1, \dots, X_n$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ random variables, with $\mu \in \mathbb{R}$ and $\sigma^2 > 0$.
 - Compute the MLE $(\hat{\mu}_{MLE}, \hat{\sigma}_{MLE}^2)$.
 - Compute the Fisher information matrix for one observation for the parameter (μ, σ^2) .
 - Deduce the asymptotic distribution of

$$\sqrt{n} \begin{pmatrix} \hat{\mu}_{MLE} - \mu \\ \hat{\sigma}_{MLE}^2 - \sigma^2 \end{pmatrix}.$$

6 Delta method and log-odds

Let $X_1, \dots, X_n$ be i.i.d. $Ber(p)$ random variables with $p \in (0, 1)$ and

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i.$$

The *odds* associated with a probability p are $p/(1-p)$ and the *log-odds* are

$$g(p) = \log \left(\frac{p}{1-p} \right).$$

- Recall the asymptotic distribution of $\sqrt{n}(\hat{p} - p)$.
- Use the delta method to obtain the asymptotic distribution of

$$\sqrt{n}(g(\hat{p}) - g(p)).$$

- In binary logistic regression, one models a binary response $Y \in \{0, 1\}$ from covariates $x \in \mathbb{R}^d$ through

$$\mathbb{P}(Y = 1 \mid X = x) = \frac{1}{1 + \exp(-x^\top \beta)}.$$

Show that the conditional log-odds are linear in x .

7 Linear regression

The multiple linear regression model is

$$Y = X\beta + \varepsilon,$$

where $Y \in \mathbb{R}^n$ is the random response vector, $X \in \mathbb{R}^{n \times p}$ is a known design matrix of rank p , $\beta \in \mathbb{R}^p$ is the unknown parameter and $\varepsilon \in \mathbb{R}^n$ is a centered random vector with covariance matrix $\sigma^2 I_n$. The least squares estimator is

$$\hat{\beta}_{LSE} = \operatorname{argmin}_{b \in \mathbb{R}^p} \|Y - Xb\|_2^2.$$

Exercise 5: Maximum likelihood estimator

$$X_i \sim J(p)$$

$$\forall p \in]0,1[, L(z) = \prod_{i=1}^n L(X_i, p) = \prod_{i=1}^n p^{X_i} (1-p)^{1-X_i}$$
$$= p^{\sum X_i} (1-p)^{n - \sum X_i}$$

$$\forall p \in]0,1[:$$

$$\underbrace{\log \circ L(z, p)}_{\ell(p)} = \log p \sum X_i + (n - \sum X_i) \log(1-p)$$

$$\ell'(p) = \frac{\sum X_i}{p} - \frac{n - \sum X_i}{1-p}$$

$$\ell'(p) = 0 \Leftrightarrow (1-p) \sum X_i = p(n - \sum X_i)$$

$$\Leftrightarrow \frac{\sum X_i}{n} = p$$

$$\text{So } \hat{p} = \frac{1}{n} \sum X_i$$

$$2) p \in]0,1[. \quad \mathbb{I}_z(p) = n \mathbb{I}_{X_1}(p)$$
$$\ell''_{X_1}(p) = \left(\frac{X_1}{p}, \frac{1-X_1}{1-p} \right)' = \left(\frac{(1-p)X_1 - p(1-X_1)}{p(1-p)} \right)'$$
$$= \left(\frac{X_1 - p}{(1-p)p} \right)' = \left(\frac{X_1 - p}{p(1-p)} \right)'$$

$$I_2(p) = n I_{X_1}(p) = n E \left[\left(\frac{X_1 - p}{p(1-p)} \right)^2 \right] \stackrel{(*)}{=} n \frac{\text{Var}(X_1)}{p^2(1-p)^2}$$

$$= \frac{1}{p(1-p)} \left(= \frac{1}{\text{Variance}} \right)$$

$$\text{CLT: } \sqrt{n} (\hat{p} - p) \longrightarrow \mathcal{N}(0, p(1-p))$$

$$\text{Var}(X) = E \left((X - E(X))^2 \right) \quad \text{(*)}$$

1. Show that

$$Y - X\hat{\beta}_{LSE}$$

is orthogonal to the column space of X . Interpret $X\hat{\beta}_{LSE}$ geometrically and deduce

$$\hat{\beta}_{LSE} = (X^\top X)^{-1} X^\top Y.$$

2. Compute the bias and covariance matrix of $\hat{\beta}_{LSE}$.

3. Assume in addition that

$$\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n).$$

Show that $\hat{\beta}_{LSE}$ is also the MLE of β when σ^2 is fixed, and give its exact distribution.

4. *Ridge regression* is the regularized estimator

$$\hat{\beta}_\lambda = \underset{b \in \mathbb{R}^p}{\operatorname{argmin}} \{ \|Y - Xb\|_2^2 + \lambda \|b\|_2^2 \}, \quad \lambda > 0.$$

Compute $\hat{\beta}_\lambda$. Is it unbiased? Explain briefly why adding bias can nevertheless reduce mean squared error.

5. Let

$$X = USV^\top$$

be an SVD of X , with non-zero singular values $s_1 \geq \dots \geq s_p > 0$. *Principal component regression with k components* means projecting the covariates on the first k right singular vectors $v_1, \dots, v_k$ and then performing ordinary least squares using only these k projected variables. Express $\hat{\beta}_{LSE}$, $\hat{\beta}_\lambda$ and the principal component regression estimator in the basis (v_j) . Explain why principal component regression is a hard spectral truncation whereas ridge regression is a soft spectral shrinkage.

8 Bayesian estimation: Beta–Bernoulli model

Let

$$X_1, \dots, X_n \mid \theta \stackrel{iid}{\sim} \operatorname{Ber}(\theta), \quad \theta \sim \operatorname{Beta}(a, b),$$

where $a, b > 0$. The prior density of θ is proportional to

$$\theta^{a-1} (1 - \theta)^{b-1} \mathbb{1}_{[0,1]}(\theta).$$

Let $S = \sum_{i=1}^n X_i$.

1. Compute the posterior distribution of θ .
2. Compute the posterior mean. Show that it is a weighted average of the prior mean and the empirical frequency S/n .
3. Compute the MAP estimator when the posterior mode lies in $(0, 1)$.
4. The *posterior predictive probability* of a new success is

$$\mathbb{P}(X_{n+1} = 1 \mid X_1, \dots, X_n).$$

Compute it.

5. For the uniform prior $a = b = 1$, show that this probability is

$$\frac{S+1}{n+2}.$$

Why can this smoothing be useful when $S = 0$ or $S = n$?

9 Conjugate Gaussian model

Assume

$$X_1, \dots, X_n \mid \mu \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2), \quad \mu \sim \mathcal{N}(\mu_0, \sigma_0^2),$$

where σ^2 is known. A family of prior distributions is called *conjugate* for a statistical model if the posterior distribution belongs to the same family as the prior.

1. Show that the posterior distribution of μ is Gaussian and compute its mean μ_1 and variance σ_1^2 .
2. Rewrite μ_1 as a weighted average of μ_0 and $\bar{X}_n$. The *precision* of a Gaussian distribution is the inverse of its variance. Interpret the weights in terms of prior and data precisions.
3. Show that the MAP estimator is equal to the posterior mean. If $\mu_0 = 0$, show that it solves

$$\operatorname{argmin}_{\mu \in \mathbb{R}} \left\{ \sum_{i=1}^n (X_i - \mu)^2 + \lambda \mu^2 \right\}$$

and identify λ .

4. Show that

$$\mu_1 = \operatorname{argmin}_{a \in \mathbb{R}} \mathbb{E} [(a - \mu)^2 \mid X_1, \dots, X_n].$$

5. Describe the behavior of the posterior mean when $n \rightarrow \infty$, when $\sigma_0^2 \rightarrow \infty$, and when $\sigma_0^2 \rightarrow 0$.

10 Confidence interval, test, power and p-value

Let $X_1, \dots, X_n$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ random variables, where $\sigma > 0$ is known. Let Φ denote the cumulative distribution function of $\mathcal{N}(0, 1)$ and let z_q be its quantile of order q :

$$\Phi(z_q) = q.$$

1. Construct an exact confidence interval for μ with coverage probability $1 - \alpha$.
2. Consider

$$H_0 : \mu = \mu_0 \quad \text{against} \quad H_1 : \mu > \mu_0.$$

Construct a test of level α based on $\bar{X}_n$.

3. The *power* at a value μ is the probability of rejecting H_0 when the true parameter is μ . Compute the power function.
4. For an observed value $\bar{x}_n$, the *p-value* is the smallest test level for which the observed statistic would lead to rejection. Compute the p-value of the previous one-sided test. Explain why it is not $\mathbb{P}(H_0 \mid \text{data})$.
5. For the two-sided test

$$H_0 : \mu = \mu_0 \quad \text{against} \quad H_1 : \mu \neq \mu_0,$$

show the duality between rejection at level α and the $(1 - \alpha)$ confidence interval.

11 Sufficient statistics

Let

$$X_1, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(p).$$

A statistic $T = T(X_1, \dots, X_n)$ is *sufficient* for p if the conditional distribution of the sample given T does not depend on p . The Fisher–Neyman factorization theorem states that, in a dominated model, T is sufficient if and only if the likelihood can be written

$$L(p; x) = h(x) g_p(T(x)).$$

1. Use the factorization theorem to show that

$$T = \sum_{i=1}^n X_i$$

is sufficient for p .

2. Show directly that the conditional distribution of $(X_1, \dots, X_n)$ given $T = k$ does not depend on p . Interpret this result as a compression of the data.
3. Let $U = X_1$. Compute

$$\mathbb{E}[U \mid T].$$

Compare its variance with that of U .

4. The Rao–Blackwell theorem states that replacing an estimator U by $\mathbb{E}[U \mid T]$, where T is sufficient, preserves its expectation and cannot increase its quadratic risk. Explain how the previous question illustrates this result.

12 AUC and the Mann–Whitney statistic

Consider a binary classification problem. A classifier assigns to each observation x a real score $s(x)$, with larger scores indicating stronger evidence for class 1. For a threshold t , define

$$TPR(t) = \mathbb{P}(s(X) > t \mid Y = 1), \quad FPR(t) = \mathbb{P}(s(X) > t \mid Y = 0).$$

The *ROC curve* plots $TPR(t)$ against $FPR(t)$ as the threshold varies, and the *AUC* is the area under this curve.

Let S_+ be the score of an independent random positive observation and S_- the score of an independent random negative observation. Assume that the score distributions are continuous, so that ties have probability zero.

1. Show that

$$AUC = \mathbb{P}(S_+ > S_-).$$

Give an interpretation of this identity.

2. Given observed positive scores $S_1^+, \dots, S_{n_+}^+$ and negative scores $S_1^-, \dots, S_{n_-}^-$, show that

$$\widehat{AUC} = \frac{1}{n_+ n_-} \sum_{i=1}^{n_+} \sum_{j=1}^{n_-} \mathbb{1}_{\{S_i^+ > S_j^-\}}.$$

3. The Mann–Whitney statistic counts the number of positive–negative pairs ordered in one direction. Explain its relation with $\widehat{AUC}$.